Designing an Intelligent Information and Analytical System for Evaluating Solutions in Renewable Energy Based on Digital Twins

Vladimir S. Simankov
Laboratory of Renewable Energy Sources
Adyghe State University
Maykop, Russia
vs@simankov.ru

Pavel Yu. Buchatskiy
Laboratory of Renewable Energy Sources
Adyghe State University
Maykop, Russia
buch@adygnet.ru

Semen V. Teploukhov
Laboratory of Renewable Energy Sources
Adyghe State University
Maykop, Russia
tepl_sv@adygnet.ru

Alexander N. Cherkasov
Laboratory of Renewable Energy Sources
Adyghe State University
Maykop, Russia
cherk@mail.ru

Stefan V. Onishchenko
Laboratory of Renewable Energy Sources
Adyghe State University
Maykop, Russia
osv@adygnet.ru

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***Abstract* — The development of the energy industry is conditioned by the main provisions of the national program "Digital Economy of the Russian Federation" and the project "Digital Energy" developed by the Ministry of Energy of Russia. The creation of digital twins is a modern trend in the development of digital technologies. The article considers the problem of designing and developing the concept of an intelligent information and analytical system for evaluating solutions in renewable energy based on digital twins. Such a complex digital solution makes it possible to form an easy-to-use virtual copy of the power system capable of reproducing the structure, condition and behavior of the original with a sufficient degree of reliability to solve the problem of involving RES in the energy balance, taking into account dynamically changing RES technologies, the potential of renewable sources in the region and various groups of criteria for the effectiveness of the system.**

Keywords* — *renewable energy sources, system analysis, intelligent analytical system, decision making, multi-criteria optimization, intelligent methods

I. INTRODUCTION

Currently, there is a trend in the world to develop all kinds of approaches to the generation of renewable energy sources and their implementation in the common existing energy systems of regions and entire states [1, 2]. Thus, there are many approaches aimed at assessing the energy potential, but the main problem is that these approaches are individual for each type of RES and do not allow for a comprehensive assessment of the effectiveness of using a combination of RES generation technologies.

In this regard, it is proposed to develop an intelligent information and analytical system that allows, at various planning levels and taking into account various criteria, to assess the potential of renewable energy sources in order to further involve them in the energy balance of the energy system, as well as to form recommendations on the possibility of using certain types of renewable energy sources and how to generate them.

II. MAIN POSITIONS

To solve the problem, it is necessary to determine a number of input parameters that significantly affect the operation of the system:

- management level [3, 4]:
  a) strategic - requires the development and accounting of technological, investment and innovative programs for the development of an object in the context of long-term planning;
  b) tactical - analysis and development of decisions based on strategic plans in the medium / short term;
  c) operational - decision-making and production and technical management at facilities of various levels;
- time interval [5]:
  a) long-term - building plans and their implementation for a long period of time (over one year), for example, the energy strategy of the Russian Federation for the period up to 2030;
  b) medium-term - planning regimes for a month, quarter, year;
  c) short-term - small time scale (up to a month), an example of a similar task is forecasting energy generation from renewable energy sources for a week ahead;
- consumers [6]:
  a) macro-region - consideration of the functioning of the energy system of the district, several regions allows you to implement joint large infrastructure projects and cooperate taking into account resource potentials;
  b) region - many regions have a number of features when involving RES in their energy systems, which imposes restrictions on the subsequent functioning of the information and analytical system;

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c) district - a separate area, which is characterized by some homogeneity of energy consumption and the similarity of other factors, for example, geographical;

d) typical object or individual consumer - a separate household, enterprise or consumer who has specific needs and energy consumption profile.

These parameters must be taken into account when solving the problem of assessing the potential of RES and the subsequent development of recommendations for building a system with renewable energy sources.

III. STAGES OF OPERATION OF THE INFORMATION-ANALYTICAL SYSTEM

Next, it is necessary to consider the main stages of the functioning of such an information-analytical system, taking into account the systematic approach. So, it is possible to allocate subsystems: monitoring; analysis of the system with RES; capacity assessments taking into account modern technologies; optimization taking into account many criteria and factors; decision-making (construction or modification of a power system with the involvement of renewable energy sources). Figure 1 shows the structural and functional diagram of the information and analytical system for assessing the potential of renewable energy.

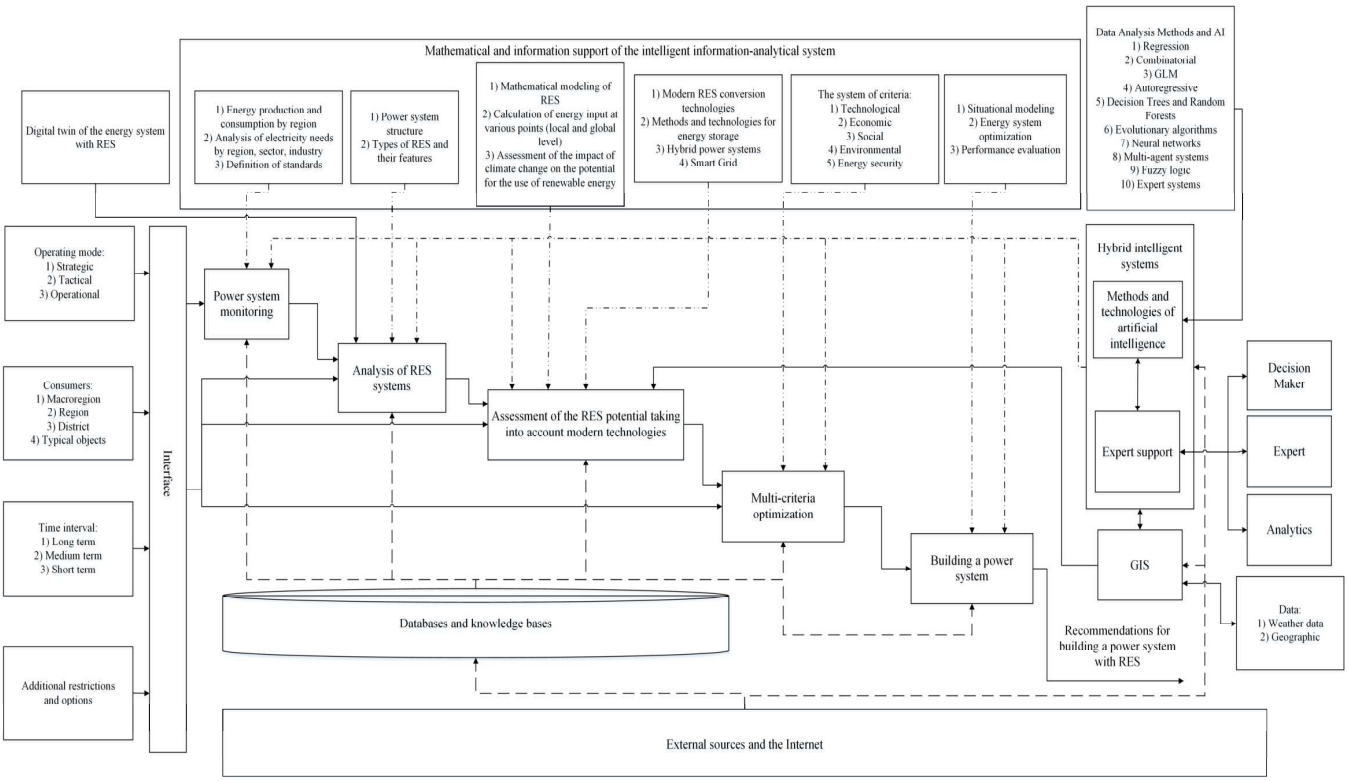


Fig. 1. Structural and functional diagram of an intelligent information and analytical system for evaluating decisions in renewable energy

Within the framework of the monitoring subsystem, the following is carried out: search and accumulation of data on the production and consumption of energy within the consumer's energy system; the needs for electricity are investigated and analyzed, taking into account the characteristics of the consumer; the standards and restrictions associated with the production and consumption of electricity are determined, for example, for an individual consumer or the wholesale and retail cost of electricity.

In the process of analyzing a system with RES, it is required to analyze the existing structure of the energy system, including both traditional and renewable sources; it is also important to identify the main types of RES that can be applied in practice, taking into account the current level of development of industry and technology.

Also, it is important to note that power systems are complex technical objects, so it is necessary to implement a digital twin subsystem to build and analyze a constantly changing digital profile of a controlled object, which is based on accumulated historical and current data, which allows you to effectively simulate the behavior of an object in the real world.

For efficient processing of multiple data streams of different nature, it is also advisable to single out a separate data analysis subsystem using AI. Within this subsystem, it is necessary to evaluate the uncertainty of the initial information [7], choose an appropriate mathematical method for data analysis using a fuzzy inference system [8], provide expert support, i.e. interaction with decision makers, experts, analysts and processing the results of their surveys. All this allows us to talk about the use of systemic (hybrid) intelligence within the IAS for making complex strategic decisions.

The next important stage in the functioning of the information and analytical system is the task of assessing the potential of renewable sources in the study area. To do this, it is necessary to take into account many factors: geographical location, meteorological features of the area, safety factors, and others. For the correct operation of this subsystem, it is necessary to perform mathematical modeling of renewable energy using complex models, which may require high-performance computing systems, it is also necessary to calculate the energy input at various points (global and local levels) for a more accurate assessment. For the effective functioning of this subsystem, it is advisable to use geographic information systems (GIS) to organize the processes of collecting, storing, visualizing and interacting with spatial data [9]. Another task to be solved in the process of assessing the potential of RES is to take into account and

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analyze modern technologies for converting and storing energy obtained from renewable sources, which requires the IAS to have a separate subsystem for interaction with external sources of information and the Internet, as well as with experts and analysts. Also important and promising is the consideration and construction of decentralized distributed systems (Smart Grid) and hybrid systems (not only autonomous complexes, but also included in a single energy system).

The next stage in the functioning of the IAS is the optimization of the studied power system according to many criteria [10, 11]. Such energy systems with RES must take into account a number of efficiency criteria (for example, economic, technical, social, safety, etc.), and it is also necessary to satisfy the demand for electricity from consumers. This allows us to conclude that there is a need for multi-criteria optimization of the studied power system.

It is also important to take into account a number of features of the functioning of power systems:

- lack of sufficient amount of retrospective information about the systems under study;
- the stochastic and often uncertain nature of the supply of renewable energy sources;
- problems of collection, processing and analysis of information on RES energy resources;
- the presence of a time delay between the receipt of information, the study of power systems with renewable energy sources and the development of decision making.

In this regard, when making multicriteria decisions, it is important to take into account the uncertainty factor of the initial information, the above features of the information, as well as the difficulty in obtaining expert information. To do this, it is necessary to determine: the numerical values of the criteria; criteria interaction functions.

The interaction function is generally described as [12]:

$$K_N = f_N(K_1, K_2, K_3, \dots, K_M) \quad (1)$$

where K_N is a criterion dependent on other criterion indicators; f_N is a criterion function depending on the criteria $K_1, K_2, K_3, \dots, K_M$.

To compare the effectiveness of a particular technology in the energy sector using renewable energy, it is necessary to take into account various groups of criteria (economic, technical, environmental, social, security, etc.). Different technologies can differ significantly in these criteria, while the combination of technologies can give a qualitatively different result, which will allow you to get a synergistic effect.

As a result, it is possible to perform simulation, situational modeling of the power system and evaluate the effectiveness, for example, evaluate the economic effect of the implemented system or reduce dependence on the supply of electricity from neighboring regions, etc.

The result of the work of the developed system are recommendations for the construction or modification of a system with RES.

IV. CONCLUSION

The paper proposes a structural and functional diagram of an intelligent information-analytical system for evaluating decisions in renewable energy. The use of digital doubles as part of such an IAS makes it possible to form a virtual copy of the power system capable of simulating the original system with a sufficient degree of reliability to solve the problem of RES involvement. Introduction to the structure of the program module of the block for accounting for the uncertainty of the initial information, the integration of the natural intelligence of the decision maker, AI methods will allow the IAS to adapt to changes in: the structure of the power system; mathematical models for assessing the potential of RES for strategic management; perform multi-criteria optimization taking into account many factors and criteria.

The platform for building an intellectual information system can be a situational center (SC), which is an intellectual complex of software and hardware for collecting, analyzing and displaying information in a form convenient for making responsible decisions. At present, sufficient experience has been accumulated in the use of SC in various fields of activity [13].

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